

Zhiyu(Joe) Wang

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WORK EXPERIENCE

Software Development Engineer - AI/ML, Adobe, *Seattle, WA*

Firefly Foundry

May 2026 - Present

- Contributed to productizing Adobe Foundry's media intelligence ingestion pipeline — implementing components of the TAC (Transcription, Analysis, Captioning) module, migrating it from a local Python prototype to the production compute platform, and integrating it into the Airflow-orchestrated video processing pipeline.
- Built an MVP agentic query layer for Adobe Foundry's media intelligence platform using LangGraph — implementing an LLM and tool loop that enables enterprise customers to conversationally query transcriptions, shot detection, and frame analysis across media libraries.

Firefly Platforms

March 2025 - May 2026

- Founding engineer on Adobe Firefly's new inference server, building a framework-agnostic alternative to an overburdened production system — designing the control plane, Redis-based request queue, and custom scale-to-zero autoscaler, integrating parallel weight download via s5cmd into a purpose-built sidecar, and leveraging filesystem snapshots to minimize cold start latency — enabling researchers to self-serve model deployments across multiple generative models on Kubernetes.
- Led end-to-end design and delivery of a GPU capacity management system for Adobe Firefly within a team of 4 — spanning a time-series forecasting pipeline for daily GPU demand prediction, an ops control plane, a self-serve UI, and a Grafana dashboard suite for monitoring, reporting, and prediction visualization — standardizing GPU resource management across the org and scaling supported model launches from 5 to 20 simultaneously.
- Standardized on-call practices for Adobe Firefly's production inference platform — building runbooks, structured incident note templates, and knowledge-sharing processes to improve response consistency and reduce tribal knowledge dependencies across the team.

Software Development Engineer, AWS, *Seattle, WA*

April 2024 - March 2025

Amazon Translate

- Led the build of core Machine translate inference image upgrading. Led the complete process of key dependencies upgrading, including Pytorch, CUDA, etc; Image tests, and production deployment.

Amazon Bedrock

- Contributed to the launch of the batch inference for Amazon Bedrock, taking ownership of key components and workflows, including batch prompt preprocessing, ensuring efficient and secure operations.
- Led the provisioned throughput metrics dashboard initiation and design. The clear and accurate data points provided by the project serves as the single sources of truth for GPU resource tracking and cost metering.

Software Development Engineer, Amazon, *Seattle, WA*

April 2021 - April 2024

- Led development of a financial data processing platform for Amazon Stores Finance — reducing processing duration from 2 hours to 5 minutes via migration to Serverless OLAP Engine, Java workflow orchestrator, and SQL modularization
- Initiated and led a data exporting feature with flexible granularity and aggregation, reducing processing time from 3 hours to under 10 minutes through Java multi-threading parallelization

Software Development Engineer, Digi International, *Boston, MA*

July 2019 - April 2021

- Contributed to Digi Accelerated Linux OS for transportation routers — router OS network features and WebUI

EDUCATION

Boston University, Boston, MA

GPA: 3.60

Sep 2017 - May 2019

Master of Science in Electrical and Computer Engineering, *Graduate with Intern Practice.*

SKILLS

Programming Languages: Python, Java, Rust, TypeScript, SQL, Bash.

Databases: PostgreSQL, Redis, DuckDB, AWS DynamoDB, AWS redshift

Technologies/Frameworks: Kubernetes, Ollama, React, MUI, S3, AWSLambda, Amazon Quicksight, Pytorch, Glue, Athena, CloudWatch, Docker, boto3, JupyterNotebook, Git.